§3 Exercises

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3.1. Tensor product of covectors
Let $e_1, \dots, e_n$ be a basis for a vector space V and let $\alpha^1, \dots, \alpha^n$ be its dual basis in $V^\vee$. Suppose $[g_{ij}] \in \mathbb{R}^{n\times n}$ is an $n \times n$ matrix. Define a bilinear function $f : V \times V \to \mathbb{R}$ by

$$f(v, w) = \sum_{1\le i,j\le n} g_{ij} v^i w^j$$

for $v = \sum v^i e_i$ and $w = \sum w^j e_j$ in V. Describe f in terms of the tensor products of α^i and α^j, $1 \le i, j \le n$.

Proof.

$$\begin{aligned} f(v, w) &= \sum_{1\le i,j\le n} g_{ij} v^i w^j \\ &= \sum_{1\le i,j\le n} g_{ij} \alpha^i(v)\alpha^j(w) \\ &= \sum_{1\le i,j\le n} g_{ij} (\alpha^i \otimes \alpha^j)(v, w). \end{aligned}$$

□

3.2. Hyperplanes

(a) Let V be a vector space of dimension n and $f : V \to \mathbb{R}$ a nonzero linear functional. Show that $\dim\ker f = n - 1$. A linear subspace of V of dimension $n - 1$ is called a *hyperplane* in V.

(b) Show that a nonzero linear functional on a vector space V is determined up to a multiplicative constant by its kernel, a hyperspace in V. In other words, if f and $g : V \to \mathbb{R}$ are nonzero linear functionals and $\ker f = \ker g$, then $g = cf$ for some constant $c \in \mathbb{R}$.

Proof. (a) Since f is nonzero, there exists $v \in V$ such that $f(v) = a \in \mathbb{R}$ nonzero. Given $c \in \mathbb{R}$, $f(\frac{c}{a}v) = \frac{c}{a}f(v) = \frac{c}{a}a = c$, so the range of f is $\mathbb{R}$. By the fundamental theorem of linear algebra,

$$\begin{aligned} n &= \dim\ker f + \dim \operatorname{range} f \\ &= \dim\ker f + 1, \end{aligned}$$

so $\dim\ker f = n - 1$.

(b) Given nonzero linear functionals $f, g : V \to \mathbb{R}$ such that $\ker f = \ker g$, let $e_1, \ldots, e_{n-1}$ be a basis of $\ker f$, and extend it to a basis $e_1, \ldots, e_{n-1}, e_n$ of V. Then $e_n \notin \ker f$ by definition, so let $r = f(e_n) \neq 0$. Similarly, let $c \in \mathbb{R}$ be such that $cg(e_n) = r$. Given any $v \in V$, write $v = \sum v^i e_i$, so that

$$\begin{aligned} f(v) &= f\left(\sum_{i=1}^{n} v^i e_i\right) \\ &= \sum_{i=1}^{n} v^i f(e_i) \\ &= v^n * r \\ &= v^n * cg(e_n) \\ &= c * (v^n e_n) \\ &= c * g(v). \end{aligned}$$

□

3.3. A basis for k-tensors
Let V be a vector space of dimension n with basis $e_1, \ldots, e_n$. Let $\alpha^1, \ldots, \alpha^n$ be the dual basis for $V^\vee$. Show that a basis for the space $L_k(V)$ of k-linear functions on V is $\{\alpha^{i_1} \otimes \cdots \otimes \alpha^{i_k}\}$ for all multi-indices $(i_1, \ldots, i_k)$ (not just the strictly ascending multi-indices as for $A_k(V)$). In particular, this shows that $\dim L_k(V) = n^k$. (This problem generalizes Problem 3.1.)

Proof. First, we show linear indepenedence. Assume $\sum c_I \alpha^I = 0$ where I ranges over all multi-indices of length k and each $c_I \in \mathbb{R}$. Apply this to e_J for some multi-index J to get **UNFINISHED**

$$0 = \sum c_I \alpha^I(e_J)$$

□

3.4. A characterization of alternating k-tensors Let f be a k-tensor on a vector space V. Prove that f is alternating if and only if f changes sign whenever two successive arguments are interchanged:

$$f(\ldots, v_{i+1}, v_i, \ldots) = -f(\ldots, v_i, v_{i+1}, \ldots)$$

for $i = 1, \ldots, k-1$.

Proof. Given a transposition $\sigma \in S_k$, $\operatorname{sgn} \sigma = -1$, so if f is alternating, by definition $f = -\sigma f$.

In the other direction, let $\sigma \in S_k$, and express σ as a product of transpositions:

$$\sigma = \sigma_1 \cdots \sigma_n.$$

Then

$$\begin{aligned}
\sigma f &= (\sigma_1 \cdots \sigma_{n-1})(\sigma_n f) \\
&= (\sigma_1 \cdots \sigma_{n-1})(-f) \\
&= (\sigma_1 \cdots \sigma_{n-2})(f) \\
&\vdots \\
&= (-1)^n f \\
&= (\operatorname{sgn} \sigma) f
\end{aligned}$$

as desired. □

3.5. Another characterization of alternating k-tensors

Let f be a k-tensor on a vector space V. Prove that f is alternating if and only if $f(v_1, \ldots, v_k) = 0$ whenever two of the vectors $v_1, \ldots, v_k$ are equal.

Proof. In the forward direction, indeed if $v_i = v_j$ for $1 \leq i < j \leq k$, then

$$f(\ldots, v_i, \ldots, v_j, \ldots) = -f(\ldots, v_j, \ldots, v_i, \ldots)$$

since swapping v_i and v_j amounts to a transposition, but in fact

$$f(\ldots, v_i, \ldots, v_j, \ldots) = f(\ldots, v_j, \ldots, v_i, \ldots)$$

because $v_i = v_j$, so

$$\begin{aligned}
f(\ldots, v_i, \ldots, v_j, \ldots) &= -f(\ldots, v_i, \ldots, v_j, \ldots) \\
\implies 2f(\ldots, v_i, \ldots, v_j, \ldots) &= 0 \\
\implies f(\ldots, v_i, \ldots, v_j, \ldots) &= 0.
\end{aligned}$$

On the other hand, if $f(v_1, \ldots, v_k) = 0$ whenever two of the vectors $v_1, \ldots, v_k$ are equal, take any $v_1, \ldots, v_k \in V$. We will proceed to show that any transposition σ yields $\sigma f = -f$ so we may apply exercise 3.4. Without loss of generality, we will assume that σ swaps v_1 and v_2 for simplicity:

$$\begin{aligned}
&f(v_1, v_2, \ldots, v_k) + f(v_2, v_1, \ldots, v_k) \\
&= f(v_1 + v_2 - v_2, v_2, \ldots, v_k) + f(v_2, v_1, \ldots, v_k) \\
&= f(v_1 + v_2, v_2, \ldots, v_k) - f(v_2, v_2, \ldots, v_k) + f(v_2, v_1, \ldots, v_k) \\
&= f(v_1 + v_2, v_2, \ldots, v_k) - 0 + f(v_2 + v_1 - v_1, v_1, \ldots, v_k) \\
&= f(v_1 + v_2, v_2, \ldots, v_k) + f(v_2 + v_1, v_1, \ldots, v_k) - f(v_1, v_1, \ldots, v_k) \\
&= f(v_1 + v_2, v_2, \ldots, v_k) + f(v_2 + v_1, v_1, \ldots, v_k) \\
&= f(v_1 + v_2, v_2 + v_1, \ldots, v_k) \\
&= 0.
\end{aligned}$$

We conclude that

$$f(v_1, v_2, \ldots, v_k) = -f(v_2, v_1, \ldots, v_k).$$

□

3.6. Wedge product and scalars

Let V be a vector space. For $a, b \in \mathbb{R}$, $f \in A_k(V)$, and $g \in A_\ell(V)$, show that $af \wedge bg = (ab)f \wedge g$.

Proof.

$$\begin{aligned}(af \wedge bg)(v_1, \ldots, v_{k+\ell}) &= \frac{1}{k!\ell!} \sum_{\sigma \in S_{k+l}} (\operatorname{sgn} \sigma) af(v_1, \ldots, v_k) * bg(v_{k+1}, v_{k+\ell}) \\ &= \frac{ab}{k!\ell!} \sum_{\sigma \in S_{k+1}} (\operatorname{sgn} \sigma)(f \otimes g)(v_1, \ldots, v_k, v_{k+1}, \ldots, v_{k+\ell}) \\ &= ab(f \wedge g)(v_1, \ldots, v_{k+\ell}).\end{aligned}$$

□

3.7. Transformation rule for a wedge product of covectors

Suppose two sets of covectors on a vector space V, $\beta^1, \ldots, \beta^k$ and $\gamma^1, \ldots, \gamma^k$, are related by

$$\beta^i = \sum_{j=1}^{k} a^i_j \gamma^j, \qquad j = 1, \ldots, k,$$

for a $k \times k$ matrix $A = [a^i_j]$. Show that

$$\beta^1 \wedge \cdots \wedge \beta^k = (\det A) \gamma^1 \wedge \cdots \wedge \gamma^k.$$

Proof.

$$\begin{aligned}\beta^1 \wedge \cdots \wedge \beta^k &= \sum_{j_1=1}^{k} a^1_{j_1} \gamma^{j_1} \wedge \cdots \wedge \sum_{j_k=1}^{k} a^k_{j_k} \gamma^{j_k} \\ &= \sum_{j_1=1}^{k} \cdots \sum_{j_k=1}^{k} a^1_{j_1} \cdots a^k_{j_k} \gamma^{j_1} \wedge \cdots \wedge \gamma^{j_k} \\ &= \sum_{\sigma \in S_k} a^1_{\sigma(1)} \cdots a^k_{\sigma(k)} \gamma^{\sigma(1)} \wedge \cdots \wedge \gamma^{\sigma(k)} \\ &= \sum_{\sigma \in S_k} a^1_{\sigma(1)} \cdots a^k_{\sigma(k)} (\operatorname{sgn} \sigma) \gamma^1 \wedge \cdots \wedge \gamma^k \\ &= (\det A) \gamma^1 \wedge \cdots \wedge \gamma^k,\end{aligned}$$

where the third equality comes from the fact that if $j_i = j_k$ for some $i \neq k$, then the wedge product is zero (Exercise 3.5). □

3.8. Transformation rule for k-covectors
Let f be a k-covector on a vector space V. Suppose two sets of vectors $u_1, \ldots, u_k$ and $v_1, \ldots, v_k$ in V are related by

$$u_j = \sum_{i=1}^{k} a_j^i v_i, \qquad j = 1, \ldots, k,$$

for a $k \times k$ matrix $A = [a_j^i]$. Show that

$$f(u_1, \ldots, u_k) = (\det A) f(v_1, \ldots, v_k).$$

Proof.

$$\begin{aligned} f(u_1, \ldots, u_k) &= f\left(\sum_{i_1=1}^{k} a_1^{i_1} v_{i_1}, \ldots, \sum_{i_k=1}^{k} a_k^{i_k} v_{i_k}\right) \\ &= \sum_{i_1=1}^{k} \cdots \sum_{i_k=1}^{k} a_1^{i_1} \cdots a_k^{i_k} f(v_{i_1}, \ldots, v_{i_k}) \\ &= \sum_{\sigma \in S_k} a_1^{\sigma(1)} \cdots a_k^{\sigma(k)} f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}) \\ &= \sum_{\sigma \in S_k} (\operatorname{sgn} \sigma) a_1^{\sigma(1)} \cdots a_k^{\sigma(k)} f(v_1, \ldots, v_k) \\ &= (\det[a_j^i]) f(v_1, \ldots, v_k), \end{aligned}$$

where the third equality comes, as in the exercise above, from the zeroing-out of terms where $i_a = i_b$ for some $a \neq b$. □

3.9. Vanishing of a covector of top degree
Let V be a vector space of dimension n. Prove that if an n-covector ω vanishes on a basis $e_1, \ldots, e_n$ for V, then ω is the zero n-covector on V.

Proof. Let $v_1, \ldots, v_n \in V$ and write $v_i = \sum_{j=1}^{n} v_i^j e_j$. Then by Exercise 3.8,

$$f(v_1, \ldots, v_n) = (\det[v_i^j]) f(e_1, \ldots, e_n) = 0.$$

□

3.10. Linear independence of covectors
Let $\alpha^1, \ldots, \alpha^k$ be 1-covectors on a vector space V. Show that $\alpha^1 \wedge \cdots \wedge \alpha^k \neq 0$ if and only if $\alpha^1, \ldots, \alpha^k$ are linearly independent in the dual space $V^\vee$.

Proof. Suppose that $\alpha^1, \ldots, \alpha^k$ are linearly *dependent*, so that there exists $\alpha_i = \sum_{j=1}^{k} c_j \alpha^j$ where $c_i = 0$. Then

$$\begin{aligned} \alpha^1 \wedge \cdots \wedge \alpha^k &= \alpha^1 \wedge \cdots \wedge \sum_{j=1}^{k} c_j \alpha^j \wedge \cdots \wedge \alpha^k \\ &= \sum_{j=1}^{k} c_j \left(\alpha^1 \wedge \cdots \wedge \alpha^j \wedge \cdots \wedge \alpha^k\right) \\ &= 0, \end{aligned}$$

since each term in line 2 is zero because either $j = i$ and thus $c_j = c_i = 0$, or $j \neq i$ and thus α^j appears twice in the wedge product (Exercise 3.5).

On the other hand, if $\alpha^1, \dots, \alpha^k$ are linearly independent, they can be extended to a basis $\alpha^1, \dots, \alpha^n$ where n is the dimension of V. Then given $e_1, \dots, e_n$, the basis dual to $\alpha^1, \dots, \alpha^n$,

$$(\alpha^1 \wedge \cdots \wedge \alpha^n)(e_1, \dots, e_n) = \det[\alpha^i(e_j)] = 1,$$

so $\alpha^1 \wedge \cdots \wedge \alpha^n$ is nonzero. □

3.11. Exterior multiplication

Let α be a nonzero 1-covector and γ a k-covector on a finite-dimensional vector space V. Show that $\alpha \wedge \gamma = 0$ if and only if $\gamma = \alpha \wedge \beta$ for some $(k-1)$-covector β on V.

Proof. Certainly, if $\gamma = \alpha \wedge \beta$, then

$$\alpha \wedge \gamma = \alpha \wedge \alpha \wedge \beta = 0$$

by Exercise 3.5.

If $\alpha \wedge \gamma = 0$, then by Exercise 3.10, α and γ are linearly dependent in $V^\vee$. UNFINISHED

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